



Graded Lab Assignment 2 – CSN344 Machine Learning, Even Sem, 2026

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Instructions

- There are 2 questions in this assignment.
- Email/paper/other modes of submissions will not be accepted.
- Upload a word **version** of this document.
- Assignment submitted after due date will not be evaluated and a score of zero will be awarded.
- Plagiarism may lead to award of zero marks.

Due Date: 1 pm, Feb., 23, 2026

Submitting this Assignment

You will submit (upload) this assignment in LMS. Name this document as GLA2_MLEVEN2026_John_Doe.doc in case your name is John Doe, and this graded lab assignment is 1 of course whose acronym is ML, and offered in EVEN 2026. Paste your code after each question, paste the screenshot of output, save and upload the document.

Grading Scheme: This assignment has 20 marks. Weight: 5 marks

Lab Question 1

Descriptive Statistics of Iris Features

Using the Iris dataset:

Tasks:

1. Compute for **Sepal Length**:
 - Variance
 - Standard Deviation
 - Skewness
 - Kurtosis
 - 25th, 50th, 75th Percentiles
 - Five-point summary

```
import numpy as np
```



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```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from scipy.stats import skew, kurtosis

iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)

sepal_length = df['sepal length (cm)']

variance = np.var(sepal_length, ddof=1)

std_dev = np.std(sepal_length, ddof=1)

skewness = skew(sepal_length)

kurt = kurtosis(sepal_length)

percentiles = np.percentile(sepal_length, [25, 50, 75])

five_point_summary = {

    "Minimum": np.min(sepal_length),

    "Q1 (25%)": percentiles[0],

    "Median (50%)": percentiles[1],

    "Q3 (75%)": percentiles[2],

    "Maximum": np.max(sepal_length)

}
```



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```
print("Variance:", variance)

print("Standard Deviation:", std_dev)

print("Skewness:", skewness)

print("Kurtosis:", kurt)

print("25th Percentile:", percentiles[0])

print("50th Percentile (Median):", percentiles[1])

print("75th Percentile:", percentiles[2])

print("Five-Point Summary:", five_point_summary)
```

OUTPUT:

```
Variance: 0.6856935123042505
Standard Deviation: 0.8280661279778629
Skewness: 0.3117530585022963
Kurtosis: -0.5735679489249765
25th Percentile: 5.1
50th Percentile (Median): 5.8
75th Percentile: 6.4
Five-Point Summary: {'Minimum': 4.3, 'Q1 (25%)': 5.1, 'Median (50%)': 5.8,
'Q3 (75%)': 6.4, 'Maximum': 7.9}
```

2. Draw:

- Histogram of Sepal Length
- Boxplot of Sepal Length

```
plt.figure(figsize=(10,4))
```

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```
plt.subplot(1,2,1)

plt.hist(sepal_length, bins=15, edgecolor='black')

plt.title("Histogram of Sepal Length")

plt.xlabel("Sepal Length (cm)")

plt.ylabel("Frequency")

plt.subplot(1,2,2)

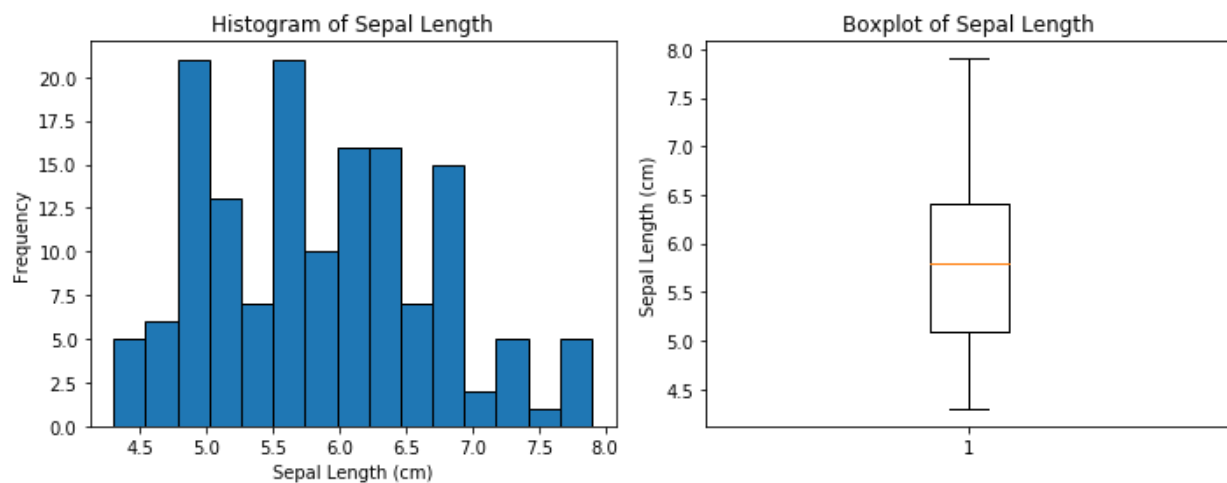
plt.boxplot(sepal_length)

plt.title("Boxplot of Sepal Length")

plt.ylabel("Sepal Length (cm)")

plt.tight_layout()

plt.show()
```



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```
3. Identify:
    o Presence of outliers
    o Shape of distribution
4. Q1 = percentiles[0]
5. Q3 = percentiles[2]
6. IQR = Q3 - Q1
7.
8. lower_bound = Q1 - 1.5 * IQR
9. upper_bound = Q3 + 1.5 * IQR
10.
11. outliers = sepal_length[(sepal_length < lower_bound) |
12.                      (sepal_length > upper_bound)]
13.
14. print("\nIQR:", IQR)
15. print("Lower Bound:", lower_bound)
16. print("Upper Bound:", upper_bound)
17.
18. if len(outliers) == 0:
19.     print("No outliers detected.")
20. else:
21.     print("Outliers detected:")
22.     print(outliers)
```

OUTPUT

```
IQR: 1.3000000000000007
Lower Bound: 3.1499999999999986
Upper Bound: 8.350000000000001
No outliers detected.
```

Total: 10 Marks

Lab Question 2

Visualization & Association Analysis

Using Iris dataset:



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Tasks:

1. Draw:

- Bar plot of Species count
- Pie chart of Species distribution
- Scatter plot of Petal Length vs Petal Width (color by Species)

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.datasets import load_iris
```

```
from scipy.stats import chi2_contingency
```

```
iris = load_iris()
```

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
```

```
df['Species'] = iris.target
```

```
df['Species'] = df['Species'].map(dict(enumerate(iris.target_names)))
```

```
df.rename(columns={
```

```
    'petal length (cm)': 'Petal_Length',
```

```
    'petal width (cm)': 'Petal_Width'
```

```
}, inplace=True)
```



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```
plt.figure(figsize=(15,4))

plt.subplot(1,3,1)

sns.countplot(x='Species', data=df)

plt.title("Bar Plot of Species Count")

plt.subplot(1,3,2)

species_counts = df['Species'].value_counts()

plt.pie(species_counts, labels=species_counts.index, autopct='%1.1f%%')

plt.title("Pie Chart of Species Distribution")

plt.subplot(1,3,3)

sns.scatterplot(x='Petal_Length', y='Petal_Width',

                hue='Species', data=df)

plt.title("Petal Length vs Petal Width")

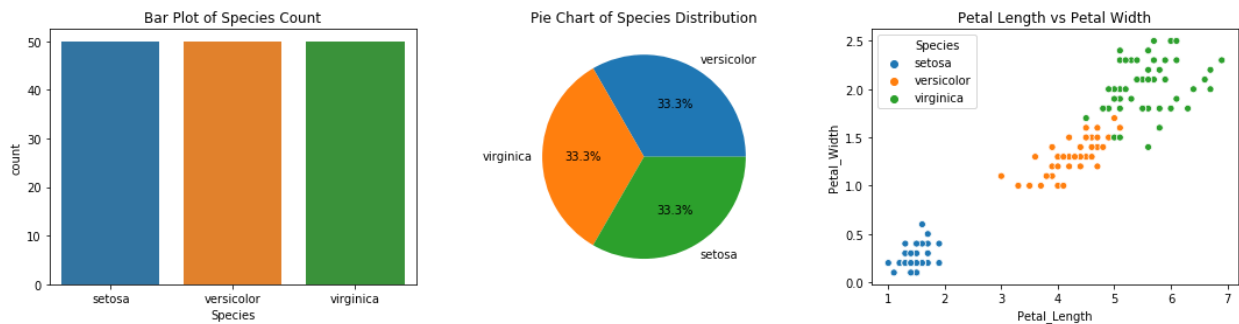
plt.tight_layout()

plt.show()
```

OUTPUT

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2. Create a Two-Way Table:

- Categorize Petal Length into:
 - Small (<2 cm)
 - Medium (2–5 cm)
 - Large (>5 cm)
- Cross-tabulate with Species

```
conditions = [
    (df['Petal_Length'] < 2),
    (df['Petal_Length'] >= 2) & (df['Petal_Length'] <= 5),
    (df['Petal_Length'] > 5)
]
```

```
categories = ['Small', 'Medium', 'Large']
```

```
df['Petal_Category'] = np.select(conditions, categories)
```

```
crosstab = pd.crosstab(df['Petal_Category'], df['Species'])
```

```
print("\nTwo-Way Table:")
```


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```
print(crosstab)
```

OUTPUT

```
Two-Way Table:
Species      setosa  versicolor  virginica
Petal_Category
Large         0           1           41
Medium        0          49           9
Small        50           0           0
```

3. Perform:

- Chi-Square Test to check association between Petal Length Category and Species
- Interpret result (p-value and conclusion)

```
chi2, p, dof, expected = chi2_contingency(crosstab)
```

```
print("\nChi-Square Statistic:", chi2)
```

```
print("Degrees of Freedom:", dof)
```

```
print("P-value:", p)
```

```
alpha = 0.05
```

```
if p < alpha:
```

```
    print("\nConclusion: Reject H0.")
```

```
    print("There is a significant association between Petal Length Category and Species.")
```

```
else:
```

```
    print("\nConclusion: Fail to Reject H0.")
```

```
    print("There is no significant association between Petal Length Category and Species.")
```



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Chi-Square Statistic: 248.5221674876847
Degrees of Freedom: 4
P-value: 1.3549185215991516e-52

Conclusion: Reject H_0 .

There is a significant association between Petal Length Category and Species.

Total: 10 Marks